

CivicPulse

Trust-Aware Civic and Market Intelligence Platform

15-Day MVP Project Plan

Final Year 6-Credit Undergraduate Project

Combines Price Patrol + GreenGuard + Trust-Aware Multimodal Report Verification

1. Supervisor-Ready Short Project Plan

Project Objective: Develop a crowdsourced urban intelligence platform that collects commodity-price observations and civic/environmental incidents, evaluates the credibility of citizen reports, and converts trusted observations into actionable geographic and temporal insights.

Core modules

Module	Main Function
Price Patrol	Monitor 3–5 essential commodities, compare markets, calculate local price statistics and detect abnormal price movements.
GreenGuard	Collect reports on waste, pollution, tree cutting, emissions, open burning, drainage/waterlogging and other civic/environmental problems.
Trust & Evidence Engine	Evaluate reporter history, evidence consistency, location consistency, community agreement and verification history.
Multimodal Verification	Compare report text with uploaded image to identify supported, weakly supported or mismatched evidence.
Duplicate Detection	Detect probable duplicate reports using location, time, category, text and image similarity.
Geospatial Intelligence	Discover trust-weighted hotspots and visualize incidents and price anomalies on an interactive map.
Emerging Issue Detection	Compare recent activity with historical/local baselines to identify rapidly increasing urban problems.
Authority Dashboard	Provide reports, alerts, maps, trends, trust information and explainable verification results.

Core technical contribution

Trust-Aware Civic Intelligence: reports are not treated as equally trustworthy. The system combines reporter reliability, evidence consistency, spatial consistency, community agreement and verification history to produce a dynamic trust score. Trusted reports contribute more strongly to hotspot and emerging-issue analysis.

MVP boundary

MUST HAVE: authentication, citizen reporting, real GPS, image upload, 3–5 commodities, price anomaly detection, image-text consistency, duplicate detection, dynamic trust score, report classification, trust-weighted hotspots, emerging issue detection, interactive map, admin dashboard and explainable decisions.

FUTURE: government API integration, mobile app, nationwide deployment, advanced multimodal models, IoT/satellite integration, advanced forecasting and agency integration.

2. Detailed Technical Plan

Recommended architecture

React + Vite Frontend → FastAPI Backend → PostgreSQL/PostGIS → AI/Analytics Engines → Authority Dashboard. Use a modular monolith for the 15-day MVP; do not build microservices.

Recommended technology stack

Layer	Technology
Frontend	React, Vite, Tailwind CSS, Axios, React Router
Maps	Leaflet / React-Leaflet, PostGIS
Backend	Python, FastAPI
Database	PostgreSQL + PostGIS
ML/NLP	Python, sentence-transformers
Computer Vision	Pretrained CLIP
Similarity	Cosine similarity
Statistics	NumPy, Pandas, SciPy
Clustering	DBSCAN
Charts	Recharts
Development	Git + GitHub, VS Code

Core data model

users; reports; price_reports; civic_reports; evidence; trust_scores; duplicates; hotspots; commodities; markets. Store latitude/longitude as PostGIS geometry so distance queries and spatial clustering can be performed efficiently.

Trust model

Trust = 0.30×Reporter Reliability + 0.25×Evidence Consistency + 0.20×Location Consistency + 0.15×Community Agreement + 0.10×Historical Verification.

Normalize to 0–100. Do not reward raw report volume; otherwise spam can increase reputation.

Price intelligence

Use local median, IQR/robust deviation and percentage deviation for the MVP. Example: $(\text{Observed Price} - \text{Local Median}) / \text{Local Median} \times 100$. Flag unusually large deviations as price anomalies.

Duplicate detection

Combine geographic distance, time difference, category match, text similarity and image similarity. A high combined similarity produces a probable duplicate rather than an automatic rejection.

Hotspot and emerging-issue intelligence

Use DBSCAN for spatial clustering. Calculate trust-weighted hotspot strength using the sum of report trust × severity. For emerging issues, compare recent 7-day activity with a local historical baseline and calculate growth percentage.

3. Complete 15-Day Execution Plan

Member A — Development: all coding, integration, AI/analytics and deployment.

Member B — Research & Documentation: literature, real-data collection, ground truth, methodology, evaluation design, report and presentation.

Day 1 — Foundation

Member A — Development: Setup React/Vite, FastAPI, PostgreSQL/PostGIS, Git repository, database schema and API structure.

Member B — Research/Documentation: Finalize problem statement/objectives/RQs; collect 10–15 relevant papers; create literature matrix.

Day 2 — Authentication & Database

Member A — Development: Implement registration, login, JWT, roles, user profile and database relationships.

Member B — Research/Documentation: Research crowdsourcing, civic reporting, VGI credibility, citizen trust and duplicate detection.

Day 3 — Citizen Reporting

Member A — Development: Build civic and price report forms with image, GPS, timestamp, category and metadata.

Member B — Research/Documentation: Begin genuine pilot-data collection; target 20–30 real reports and document collection context.

Day 4 — Price Patrol

Member A — Development: Implement commodity/market APIs, price submission, history, local statistics and comparison.

Member B — Research/Documentation: Collect real prices for rice, onion, potato, soybean oil and egg; document source, market and date.

Day 5 — Price Anomaly

Member A — Development: Implement rolling median, IQR/deviation calculations, anomaly flags and price charts.

Member B — Research/Documentation: Research statistical anomaly detection and document the mathematical rationale.

Day 6 — GreenGuard

Member A — Development: Implement all selected civic/environmental categories in the unified reporting engine.

Member B — Research/Documentation: Collect 30–50 genuine civic/environmental observations and document evidence.

Day 7 — Image/Text Intelligence

Member A — Development: Integrate pretrained CLIP; generate image/text embeddings and cosine-similarity consistency score.

Member B — Research/Documentation: Prepare matching, mismatching and ambiguous real examples; document evaluation methodology.

Day 8 — Duplicate Detection

Member A — Development: Implement geo + time + category + text + image similarity and probable-duplicate detection.

Member B — Research/Documentation: Review duplicate-report literature; prepare ground-truth duplicate cases where available.

Day 9 — Trust Engine

Member A — Development: Implement reporter reliability, evidence consistency, location consistency, community agreement and historical verification.

Member B — Research/Documentation: Research credibility/trust models; justify every trust variable and weighting choice.

Day 10 — Verification Engine

Member A — Development: Fuse trust, evidence, duplicate and consistency signals into supported/suspicious/duplicate/low-credibility outputs with reasons.

Member B — Research/Documentation: Define labels and evaluation methodology; prepare confusion-matrix and error-analysis plan.

Day 11 — GIS Intelligence

Member A — Development: Implement PostGIS spatial queries, Leaflet maps, DBSCAN clustering and trust-weighted hotspots.

Member B — Research/Documentation: Document GIS methodology, clustering assumptions and hotspot definition.

Day 12 — Emerging Issues

Member A — Development: Implement 7-day trend comparison, growth percentage, trust-weighted activity and alerts.

Member B — Research/Documentation: Research temporal/emerging-issue detection; prepare result figures and methodology.

Day 13 — Dashboard

Member A — Development: Build citizen dashboard and authority dashboard: KPIs, maps, charts, alerts, trust and verification details.

Member B — Research/Documentation: Prepare architecture/database/workflow figures and begin final documentation.

Day 14 — Integration & Testing

Member A — Development: End-to-end testing: auth, reports, images, GPS, price, AI, trust, duplicates, hotspots and dashboard.

Member B — Research/Documentation: Create test cases, expected/actual results and document failures, fixes and limitations.

Day 15 — Evaluation & Freeze

Member A — Development: Run final evaluation, generate metrics, freeze MVP, prepare demo data and final presentation environment.

Member B — Research/Documentation: Finish abstract, introduction, literature, methodology, results, limitations, future work, references and viva questions.

4. Final Demo Flow

1. Citizen logs in.
2. Submits a price or civic/environmental report with text, image and location.
3. Backend preprocesses the evidence.
4. Image-text consistency is calculated.
5. Duplicate candidates are checked.
6. Reporter trust is calculated.
7. Report receives an explainable credibility status.
8. Price reports feed anomaly analysis; civic reports feed risk/hotspot analysis.
9. Trust-weighted spatial clusters and emerging issues appear on the authority dashboard.
10. Administrator can review, assign, update and resolve reported incidents.

5. Project Positioning

Do not present this as three separate projects. Price Patrol and GreenGuard are the two application domains.

Trust-Aware Multimodal Verification is the shared technical intelligence/security layer. The central research/engineering question is whether heterogeneous crowdsourced observations can be made more useful for urban decision-making by estimating report credibility before using them for price anomalies, hotspots and emerging-issue detection.

6. Important Scientific Boundary

The system should not claim to prove that a citizen's allegation is legally true or false. Use terms such as **supported, suspicious, duplicate, low-credibility, or requires human verification**. The platform is a decision-support system; final enforcement decisions remain with authorized authorities.

Prepared as the consolidated 15-day MVP plan for the CivicPulse final-year project.